# SAI KUMAR BYLAPUDI

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**https://sk-portfolio-lemon.vercel.app**

## SUMMARY

MSc Data Science student with a strong background in computer science. I enjoy bringing ideas to life—whether building software, analysing data, or applying machine learning. I have worked with Python, Java, SQL, FastAPI, React, and cloud-based tools, I'm passionate about solving real-world problems through creativity and collaboration, and constantly exploring emerging technologies in cloud computing, AI, and analytics to drive meaningful impact.

## EDUCATION

**University of Hertfordshire** 2024 - 2026

MSc in Data Science (with placement Year)
Modules: Data Handling and Visualisation, Machine Learning

**Jawaharlal Nehru Technological university** 2019 - 2023

B. Tech in Computer Science (First Class)
Modules: Java, Data Structures, SQL, Artificial intelligence

## WORK EXPERIENCE

**Regent home care Ltd** Nov 2024 - Present

**Domicile Carer**

- Supporting service users with daily care needs while maintaining dignity, privacy, and confidentiality.
- Communicate clearly with service users, families, and team members to provide safe and effective support.
- Follow care plans, record information accurately, and report concerns when required.
- Developed strong communication, empathy, teamwork, and time-management skills in a people-focused environment.

**Mthree** Apr 2024 - Jun 2024

**Software Developer Intern**

- Collaborated within a 3-member Agile team, delivering a Java-based banking project **2 weeks ahead of the planned schedule**.
- Improved software performance by **15%** through efficient implementation of Data Structures and Spring Boot optimization.
- Applied AWS cloud and SQL database solutions, reducing data retrieval latency by **20%** and enhancing application responsiveness.

**Hexaware** Apr 2022 - Aug 2022

**Graduate Engineer trainee Apprentice**

- Optimized SQL queries, achieving a **25% increase** in data processing speed and reducing server load.
- Debugged and resolved **over 50 code issues** during apprenticeship, significantly enhancing code quality and reliability.
- Engaged in hands-on coding exercises, delivering clean and maintainable Java code for **5 major feature deployments** during training.

## SKILLS

**Programming Languages:** C++, JavaScript, SQL
**Tools and IDEs:** Spring Tool Suite, IntelliJ IDEA, Eclipse, PyCharm, Git (Version control), VS Code

## Certifications and Simulations

- **Microsoft** Azure AI Fundamentals (AI900)
- **Microsoft** Azure Data Fundamentals (DP900)
- **IBM** SQL and Relational Databases 101
- **Mthree** Software Developer
- **JP Morgan Chase & Co** Software Engineering Virtual Job Simulation
- **Accenture, UK** Introduction to Accenture Virtual Job Simulation

## Hands-on Technical Projects

### Heart Disease Prediction Application using XGBoost

*Python, Scikit-learn, XGBoost, SMOTE, StandardScaler, Streamlit*

- Developed an XGBoost model to predict heart disease risk with 85% accuracy.
- Enhanced model reliability using SMOTE for class balancing and StandardScaler for feature scaling.
- Deployed via Streamlit with a user-friendly interface for real-time input and risk classification.
- Enabled early detection support for individuals and healthcare professionals through risk-level insights (Low, Moderate, High, Critical).

### Graduate Job Match AI

*Python, FastAPI, React/Next.js, PostgreSQL, pgvector, OpenAI Embeddings*

- Developed an AI-based platform that compares graduate CVs with job descriptions and provides a match score.
- Used OpenAI embeddings and pgvector for semantic matching between candidate skills and job requirements.
- Built FastAPI endpoints for CV upload, job analysis, skill-gap detection, and interview preparation suggestions.
- Designed the project to support graduates in improving applications and preparing more effectively for roles.
- Used Git-based version control and structured the project for future deployment

## Additional Information

- **Hackathon Coordinator**: Led an 8-member team to organize a college hackathon.
- **Event Coordination:** Managed college tech and cultural fests.
- **Club Involvement:** Active member of dance and singing clubs.

## Hobbies and Interests

- Passionate about **fitness and self-discipline**, with regular **morning gym sessions**.
- Actively improve my **coding and problem-solving skills** by solving challenges on **LeetCode**.
- Enjoy reading **personality development books** that strengthen focus and determination.
- Love **exploring new places** through trekking and unwinding with peaceful **beach sunsets**.

## References

- Flexible to work on graduate route visa if the role doesn't meet the immigration criteria.